GitHub link:

[TanviUttla06/MLOps-Tanvi-B23BB1044]

Q1:

Assignment 1.ipynb
assignment_1_2nd half.ipynb

Epoch : 5

Pin_Memory : True

MNIST:

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	99.10%	98.68%
16	**SGD**	**0.0001**	98.56%	97.74%
16	**Adam**	**0.001**	98.67%	98.60%
16	**Adam**	**0.0001**	98.71%	97.96%
32	**SGD**	**0.001**	98.96%	98.72%
32 [Optional]	**SGD**	**0.0001**	-	-
32 [Optional]	**Adam**	**0.001**	-	-
32	**Adam**	**0.0001**	98.74%	97.03%

FashionMNIST:

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	91.02%	88.81%
16	**SGD**	**0.0001**	88.29%	84.85%
16	**Adam**	**0.001**	90.72%	87.96%
16	**Adam**	**0.0001**	89.94%	88.23%
32	**SGD**	**0.001**	90.19%	88.61%
32 [Optional]	**SGD**	**0.0001**	-	-
32 [Optional]	**Adam**	**0.001**	-	-
32	**Adam**	**0.0001**	89.83%	87.14%

Pin_Memory : False

MNIST:

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	98.96%	98.57%

16	**SGD**	**0.0001**	98.43%	97.94%
16	**Adam**	**0.001**	98.81%	98.40%
16	**Adam**	**0.0001**	98.49%	98.33%
32	**SGD**	**0.001**	98.95%	98.60%
32 [Optional]	**SGD**	**0.0001**	-	-
32 [Optional]	**Adam**	**0.001**	-	-
32	**Adam**	**0.0001**	98.49%	97.43%

FashionMNIST:

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	90.07%	89.01%
16	**SGD**	**0.0001**	89.05%	84.59%
16	**Adam**	**0.001**	90.68%	85.12%
16	**Adam**	**0.0001**	90.20%	88.10%
32	**SGD**	**0.001**	90.40%	87.25%
32 [Optional]	**SGD**	**0.0001**	-	-
32 [Optional]	**Adam**	**0.001**	-	-
32	**Adam**	**0.0001**	89.84%	87.50%

Epoch : 2

Pin_Memory: True

MNIST:

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	98.66%	98.13%
16	**SGD**	**0.0001**	97.96%	96.39%
16	**Adam**	**0.001**	98.59%	97.27%
16	**Adam**	**0.0001**	98.74%	96.79%
32	**SGD**	**0.001**	98.54%	97.67%
32 [Optional]	**SGD**	**0.0001**	-	-
32 [Optional]	**Adam**	**0.001**	-	-
32	**Adam**	**0.0001**	98.28%	95.52%

FashionMNIST:

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	88.34%	86.33%
16	**SGD**	**0.0001**	86.78%	81.93%
16	**Adam**	**0.001**	89.19%	83.30%
16	**Adam**	**0.0001**	89.11%	84.67%

32	**SGD**	**0.001**	88.58%	85.65%
32 [Optional]	**SGD**	**0.0001**	-	-
32 [Optional]	**Adam**	**0.001**	-	-
32	**Adam**	**0.0001**	88.86%	83.64%

Pin_Memory: False

MNIST:

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	98.92%	98.18%
16	**SGD**	**0.0001**	97.81%	95.91%
16	**Adam**	**0.001**	98.51%	96.92%
16	**Adam**	**0.0001**	98.51%	96.19%
32	**SGD**	**0.001**	98.69%	97.64%
32 [Optional]	**SGD**	**0.0001**	-	-
32 [Optional]	**Adam**	**0.001**	-	-
32	**Adam**	**0.0001**	98.45%	95.79%

FashionMNIST:

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**

16	**SGD**	**0.001**	88.51%	85.83%
16	**SGD**	**0.0001**	87.42%	81.04%
16	**Adam**	**0.001**	88.70%	67.82%
16	**Adam**	**0.0001**	88.51%	84.78%
32	**SGD**	**0.001**	88.95%	85.29%
32 [Optional]	**SGD**	**0.0001**	-	-
32 [Optional]	**Adam**	**0.001**	-	-
32	**Adam**	**0.0001**	88.63%	83.89%

🧠 Detailed Analysis

1. ResNet-18 vs. ResNet-50

Counter-intuitively, **ResNet-18 consistently outperformed ResNet-50** across almost all tests. This is likely due to:

- **Dataset Complexity:** MNIST and FashionMNIST are relatively simple (28x28 grayscale). The higher capacity of ResNet-50 (Bottleneck blocks) likely leads to overfitting or requires more epochs to converge from scratch compared to the shallower ResNet-18.
- **Vanishing Gradients:** Without pre-training, deeper models like ResNet-50 are harder to optimize in very few epochs (2-5).

2. Impact of Optimizers and Learning Rate

- **SGD vs. Adam:** SGD with a Learning Rate of 0.001 generally provided the highest peak accuracy for MNIST (99.10%). Adam was more stable across different learning rates but occasionally lagged behind SGD in final percentage points on simpler datasets.
- **Learning Rate Sensitivity:** A LR of 0.001 was the "sweet spot." Dropping to 0.0001 resulted in a noticeable dip in accuracy, particularly for the deeper ResNet-50, which requires higher gradients to update its many layers early on.

3. Pin_Memory and Epochs

- **Pin_Memory:** While accuracy remains largely unchanged between `True` and `False`, `Pin_Memory: True` showed faster data transfer speeds between CPU and GPU, reducing overall training time.
- **Epoch Scaling:** Increasing epochs from 2 to 5 provided a significant boost in FashionMNIST accuracy (~2-3% gain), indicating that the model had not yet reached its plateau at 2 epochs.

4. Dataset Difficulty

MNIST accuracy stayed consistently above 95%, while FashionMNIST proved more challenging, hovering between 84% and 91%. This validates that spatial features in clothing (FashionMNIST) are more complex than the strokes in digits (MNIST).

Q1.b)

```
=== FINAL SUMMARY TABLE FOR Q1(b) ===
Dataset          | Kernel      | Accuracy    | Time (ms)
-------------------------------------------------------
MNIST            | poly        | 95.15%      | 8372
MNIST            | rbf         | 95.94%      | 8789
FashionMNIST     | poly        | 81.66%      | 8664
FashionMNIST     | rbf         | 85.31%      | 9088
```

1. Kernel Performance Comparison

- **RBF (Radial Basis Function) Kernel:** Consistently outperformed the Polynomial kernel on both datasets. For MNIST, it reached **95.94%**, and for FashionMNIST, it reached **85.31%**. This suggests that the RBF kernel's ability to map data into an infinite-dimensional space is better suited for the high-dimensional pixel data of these datasets.
- **Polynomial Kernel:** Performed reasonably well on digits (MNIST) but struggled significantly with the complex textures of clothing in FashionMNIST, trailing by nearly **4%** compared to RBF.

2. Computational Efficiency (Training Time)

- There is a direct trade-off between accuracy and training time. The **RBF kernel** required more time (approx. **9088 ms** for FashionMNIST) compared to the Polynomial kernel (**8664 ms**).
- The increased time for RBF is due to the complexity of calculating the Gaussian distance for every pair of points in the high-dimensional space.

3. SVM vs. ResNet (Comparison with Q1.a)

- **Performance Gap:** While SVM achieved respectable results (~95% on MNIST), it falls short of the **ResNet-18** performance reported in Q1(a) (**99.10%**).
- **Feature Extraction:** Unlike ResNet, which uses convolutional layers to automatically learn spatial hierarchies and features, SVM relies on raw flattened pixel values. This explains why SVM performance drops more drastically on **FashionMNIST** (where spatial texture is key) than on the simpler MNIST digits.

Q2:

Tanvi_MLops Q2.ipynb

GPU:

```
================================================================================
FINAL GPU REPORT: 4 ROWS | 6 COLUMNS
================================================================================
   Model Optimizer  GFLOPs  Params(M)  Time(s)  Accuracy(%)
RESNET18      Adam  0.0356      11.18    22.93        87.27
RESNET18       SGD  0.0356      11.18    22.07        86.35
RESNET50      Adam  0.0827      23.52    47.62        84.25
RESNET50       SGD  0.0827      23.52    44.83        83.33
```

Attaching just because:

Assignment1_Q3.ipynb :: For resnet18

Tanvi_MLops Q2.ipynb : For resnet50

Report: Quantitative Analysis of CNN Architectures and Optimizers

1. Project Objective

This study evaluates the performance of **ResNet18** and **ResNet50** architectures using **Adam** and **SGD** optimizers. We analyze how architectural depth and weight-update strategies impact computational cost (FLOPs), training latency (Time), and model accuracy on the **FashionMNIST** dataset.

2. Methodology & Parameters

The experiment is conducted under a controlled GPU environment to ensure consistency.

- **Dataset:** FashionMNIST (Grayscale, 28×28 pixels resized to 32×32).
- **Hardware:** NVIDIA CUDA-enabled GPU.
- **Metrics:** We measure 6 distinct columns: Model Type, Optimizer Type, GFLOPs (theoretical complexity), Params (memory footprint), Time (real-world latency), and Test Accuracy.

3. Comparative Architecture Analysis

We compare two tiers of Residual Networks:

- **ResNet18:** Employs **Basic Blocks**, consisting of two 3×3 convolutional layers.
- **ResNet50:** Employs **Bottleneck Blocks**, utilizing $1 \times 1 \rightarrow 3 \times 3 \rightarrow 1 \times 1$ convolutions to manage computational load across 50 layers.

4. Optimization Dynamics

- **Adam (Adaptive Moment Estimation):** Uses adaptive learning rates for each parameter by tracking the first and second moments of gradients.
- **SGD (Stochastic Gradient Descent):** Uses a fixed global learning rate with momentum to overcome local minima.

5. Discussion & Conclusion

The results highlight a critical "Efficiency Frontier" in Deep Learning:

1. **Complexity vs. Return:** ResNet50 offers significantly higher parameter counts and GFLOPs, yet for simple datasets like FashionMNIST, the accuracy gains over ResNet18 are often marginal.
2. **Hardware Latency:** SGD often demonstrates lower training time per epoch due to its reduced mathematical complexity compared to the adaptive calculations required by Adam.

3. **Summary:** For small-scale image recognition, **ResNet18 with Adam** provides the most efficient balance, achieving high accuracy with minimal computational overhead. ResNet50 is better suited for high-resolution, complex datasets where its depth can be fully utilized.